Event-level evaluation of Sentinel-1 flood segmentation on Sen1Floods11: a pre-event reference and land-cover context reduce false positives on unseen floods

22PW08 Bharath Pranav S | Code, logs and results: https://github.com/Hirata-Origami/FloodSegmentation

Abstract

Deep-learning flood mapping from Sentinel-1 radar is usually reported on the official Sen1Floods11 test split, where chips from the same flood events appear in training. We re-evaluate single-date Sentinel-1 U-Net models with leave-one-event-out (LOEO) testing over the 11 hand-labelled events, using three training seeds, paired bootstrap intervals, matched-recall precision comparisons and per-event precision. Single-date Sentinel-1 ensembles reach 0.69 water IoU on the official test split, but only 0.56-0.57 macro IoU on unseen events; the hardest events (Somalia, Pakistan) are dominated by false positives, with precision of about 0.31 and 0.38. Terrain inputs (JRC, elevation, slope, HAND), weak-label pretraining, Tversky and Lovász losses, augmentation, larger context, connected-component cleanup, land-cover or terrain exclusion masks, and event-level weather scalars gave no reliable gain, and the weather scalars, which take roughly one value per event, reduced LOEO IoU by 0.06. Supplying a same-orbit pre-event reference (one-year median, as a difference and z-score) raised LOEO macro IoU by 0.037 (95% CI 0.019 to 0.058; positive for all three seeds), Dynamic World land-cover probabilities raised it by 0.022 (0.001 to 0.043), and both together by 0.042 (0.020 to 0.064; 0.564 to 0.606). The pre-event reference accounts for most of the IoU gain, whereas land cover mainly adds precision: macro precision of seed-ensembled predictions rose from 0.71 to 0.77 (reference), 0.79 (land cover) and 0.81 (both), with Somalia precision rising from 0.31 to 0.70-0.82 and Pakistan from 0.38 to 0.59-0.74; the precision gains persist at matched recall. On the official split, 15-model ensembles improved from 0.690 to 0.706-0.710 test IoU, but the paired differences are not statistically significant on its 90 test chips. These inputs make the task multi-source and bi-temporal, so the results are not directly comparable to single-date published scores. We release all code and logs.

Leave-one-event-out (3 seeds)	macro IoU	paired diff vs S1 (95% CI)	macro precision	Somalia / Pakistan precision
S1 baseline (single date)	0.564	-	0.706	0.31 / 0.38
+ pre-event reference	0.602	+0.037 (+0.019, +0.058)	0.768	0.70 / 0.59
+ land cover (Dynamic World)	0.585	+0.022 (+0.001, +0.043)	0.790	0.82 / 0.72
+ reference + land cover	0.606	+0.042 (+0.020, +0.064)	0.805	0.81 / 0.74

Official split (15-model ensembles)	test IoU (95% CI)	paired diff vs S1 (95% CI)	Bolivia IoU
S1 baseline	0.690 (0.578-0.774)	-	0.606
+ reference	0.710 (0.612-0.787)	+0.020 (-0.001, +0.053)	0.691
+ land cover	0.710 (0.608-0.785)	+0.020 (-0.014, +0.063)	0.601
+ reference + land cover	0.706 (0.600-0.786)	+0.016 (-0.036, +0.072)	0.715

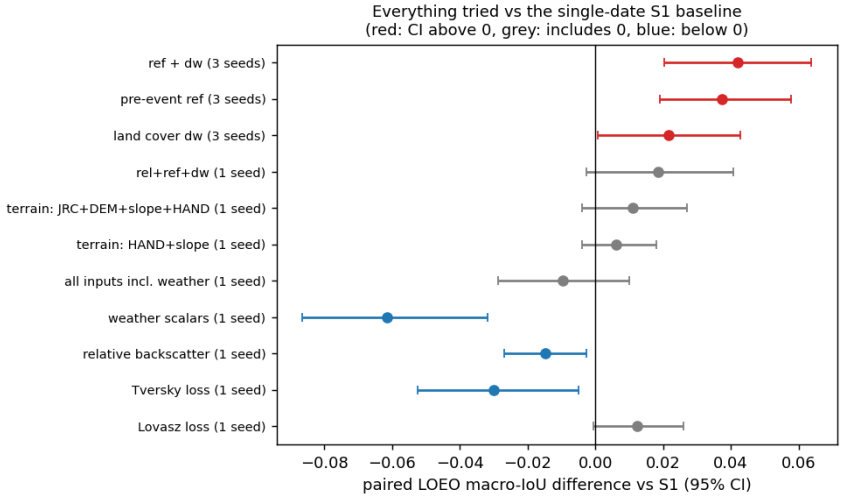


Figure. Every idea tested, as the paired leave-one-event-out macro-IoU difference vs the single-date baseline (95% CI). Only the pre-event reference, land cover and their combination clear zero.

Keywords: Sentinel-1, flood mapping, Sen1Floods11, leave-one-event-out, false positives, pre-event reference, Dynamic World